says that you can move energy around and change its form but you can't create or destroy it. Feynman makes this law crystal clear with his amusing story of Dennis the Menace who is always mischievously hiding his toy building blocks from his mother (Chapter Four).

The most challenging lecture in this volume is the last, which is an exposition on quantum physics. It is no exaggeration to say that quantum mechanics had dominated twentieth-century physics and is far and away the most successful scientific theory in existence. It is indispensable for understanding subatomic particles, atoms and nuclei, molecules and chemical bonding, the structure of solids, superconductors and superfluids, the electrical and thermal conductivity of metals and semiconductors, the structure of stars, and much else. It has practical applications ranging from the laser to the microchip. All this from a theory that at first sight—and second sight—looks absolutely crazy! Niels Bohr, one of the founders of quantum mechanics, once remarked that anybody who is not shocked by the theory hasn't understood it.

The problem is that quantum ideas strike at the very heart of what we might call commonsense reality. In particular, the idea that physical objects such as electrons or atoms enjoy an independent existence, with a complete set of physical properties at all times, is called into question. For example, an electron cannot have a position in space and a well-defined speed at the same moment. If you look for where an electron is located, you will find it at a place, and if you measure its speed you will obtain a definite answer, but you cannot make both observations at once. Nor is it meaningful to attribute definite yet unknown values for the position and speed to an electron in the absence of a complete set of observations.

This indeterminism in the very nature of atomic particles is encapsulated by Heisenberg's celebrated uncertainty principle. This puts strict limits on the precision with which properties such as position and speed can be simultaneously known. A sharp value for position smears the range of possible values of speed and vice versa. Quantum fuzziness shows up in the way electrons, photons, and other particles move. Certain experiments can reveal them taking definite paths through space, after the fashion of bullets following trajectories toward a target. But other experimental arrangements reveal that these entities can also behave like waves, showing characteristic patterns of diffraction and interference.

Feynman's masterly analysis of the famous "two-slit" experiment, which teases out the "shocking" wave-particle duality in its starkest form, has become a classic in the history of scientific exposition. With a few very simple ideas, Feynman manages to take the reader to the very heart of the